SIMATS SCHOOL OF ENGINEERING

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Assignment 5

1. **Write a finite-state automaton for a dialogue manager for checking your bank balance and withdrawing money at an automated teller machine.** (10M) (CO5) (BLT4)
2. **Consider the sentence "Christ opened the eyes", describes fact about English syntax.**
   a) Describe and illustrate how your favourite foreign language differs in this respect.
   b) Explain how a MT system could deal with this difference. (10M) (CO5) (BLT4)

Answer 1: Finite-state automaton for an ATM dialogue manager

A dialogue manager decides what the system says next from what the user has said so far. For a small, tightly controlled task such as an automated teller machine (ATM), it can be modelled as a **finite-state automaton (FSA)**: each state is one system prompt or action, each arc is labelled with the user input (or event) allowed in that state, and the machine follows exactly one arc per turn. This is a "system-initiative" dialogue: the machine asks one question at a time and the user can only answer that question.

Formal definition. The dialogue manager is the 5-tuple $M = (Q, \Sigma, \delta, q_0, F)$:

- **Q** = {q0 Idle, q1 Ask PIN, q2 Main menu, q3 Show balance, q4 Ask amount, q5 Validate amount, q6 Confirm amount, q7 Dispense cash, q8 Eject card, q9 Card retained}.
- **Σ** = {card, pin_ok, pin_bad, pin_bad3, balance, withdraw, amount, amount_ok, amount_bad, yes, no, done, cancel, timeout}; each symbol is the *dialogue act* produced by the card reader, keypad or speech recogniser.
- **δ** : $Q \times \Sigma \rightarrow Q$ is the transition function (Figure 1 and Table 2). An input that is not legal in the current state is treated as a mis-recognition: the prompt is repeated and the state does not change.
- $\mathbf{q_0}$ = Idle (start state); **F** = {q8, q9}: card ejected, or card retained after three wrong PINs.

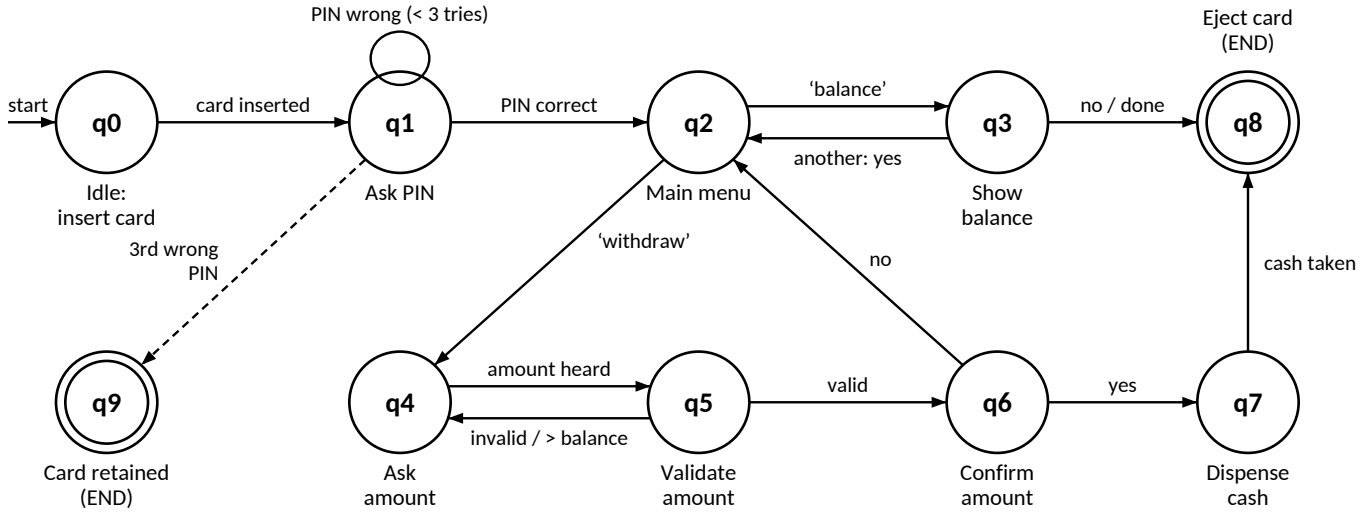
Table 1: States and the system action in each state

State	Name	System prompt / action
q0	Idle	"Please insert your card."
q1	Ask PIN	"Please enter your 4-digit PIN." (a counter records the number of tries)
q2	Main menu	"Would you like to check your balance or withdraw cash?"
q3	Show balance	Reads the balance aloud / on screen, then asks "Anything else?"
q4	Ask amount	"How much would you like to withdraw?"
q5	Validate amount	Check: multiple of note value, within daily limit, ≤ balance
q6	Confirm amount	"You asked for ₹X. Is that correct?"
q7	Dispense cash	Counts out the notes, prints the receipt, asks "Anything else?"
q8	Eject card (final)	"Please take your card. Thank you." – session ends
q9	Card retained (final)	"Your card has been retained for security." – session ends

Table 2: Transition function δ (legal moves only)

State	Input	Next	State	Input	Next
q0	card	q1	q4	amount	q5
q1	pin_ok	q2	q5	amount_ok	q6
q1	pin_bad (< 3)	q1	q5	amount_bad	q4
q1	pin_bad3	q9	q6	yes	q7
q2	balance	q3	q6	no	q2
q2	withdraw	q4	q7	done	q8
q3	yes (another)	q2	any	cancel / timeout	q8
q3	no / done	q8			

Figure 1: State-transition diagram of the ATM dialogue manager



Any state: 'cancel' / timeout → q8 (eject card). Double circle = accepting (final) state.

Circles are states, arrows are transitions labelled with the input that triggers them; the dashed arrow is the failure path after three wrong PINs.

Sample dialogue (a): withdrawal with two corrections – q0 → q1 → q1 → q2 → q4 → q5 → q4 → q5 → q6 → q7 → q8

System (q0): Please insert your card. **User:** [inserts card]

System (q1): Please enter your PIN. **User:** 4 1 2 7 (wrong)

System (q1): Incorrect PIN. Please try again. **User:** 4 1 2 9 (correct)

System (q2): Check your balance or withdraw cash? **User:** Withdraw.

System (q4): How much would you like to withdraw? **User:** ₹ 25,000.

System (q5→q4): Sorry, the daily limit is ₹ 20,000. How much? **User:** ₹ 5,000.

System (q6): You asked for ₹ 5,000. Is that correct? **User:** Yes.

System (q7): Please take your cash and receipt. **User:** [done]

System (q8): Please take your card. Thank you.

Sample dialogue (b): balance enquiry – q0 → q1 → q2 → q3 → q8

System (q3): Your balance is ₹ 48,250. Anything else? **User:** No, thanks. **System (q8):** Please take your card.

Implementation sketch. The automaton is just a lookup table plus a loop, which makes it easy to test and to audit:

```
delta = {('q0','card'):'q1',
         ('q1','pin_ok'):'q2', ('q1','pin_bad'):'q1', ('q1','pin_bad3'):'q9',
         ('q2','balance'):'q3', ('q2','withdraw'):'q4',
         ('q3','another'):'q2', ('q3','done'):'q8',
         ('q4','amount'):'q5',
         ('q5','amount_ok'):'q6', ('q5','amount_bad'):'q4',
         ('q6','yes'):'q7', ('q6','no'):'q2', ('q7','done'):'q8'}
FINAL, state = {'q8','q9'}, 'q0'
while state not in FINAL:
    act = get_input(state)                # keypad / speech act
    if act in ('cancel', 'timeout'): state = 'q8'
    else: state = delta.get((state, act), state)   # unknown input: re-prompt
    speak(prompt_for[state])
```

Design notes.

- **Verification first:** the only arc out of q0 leads to q1, so no account data is reachable before the PIN is accepted; three failures end in q9.
- **Confirmation (q6):** speech or keypad input may be wrong, so the amount is confirmed before any cash moves; invalid amounts return to q4 instead of ending the session.
- **Limits of an FSA:** the dialogue is system-initiative and rigid (the user cannot say "withdraw 5,000 rupees" in one utterance). Frame-based or information-state managers are more flexible, but for a safety-critical ATM the predictability of an FSA is an advantage.

Answer 2: "Christ opened the eyes" – a fact about English syntax

The fact. English is an **SVO** language with **fixed word order**. In "Christ opened the eyes" the subject *Christ* comes before the verb *opened* and the object *the eyes* comes after it. English nouns carry no case endings, so grammatical roles are shown *only by position*: "The eyes opened Christ" has a different (odd) meaning. The sentence also shows that the determiner *the* precedes its noun and that the verb is marked only for tense (*-ed*), with no gender or number agreement: VP → V NP.

(a) How Hindi (my chosen foreign language) differs

Hindi is an **SOV**, head-final language. The same sentence, romanised and glossed word by word:

Hindi (romanised)	**Isa Masih**	**ne**	**ānkhen**	**kholiin**
Gloss	Christ	ERG (case marker)	eyes (fem. plural)	opened (perfective, fem. plural)
English	Christ	–	the eyes	opened

- **Verb-final order.** The verb comes last (S–O–V): VP → NP V in Hindi but VP → V NP in English (Figure 2).
- **Case marking instead of position.** With a perfective transitive verb the subject takes the ergative postposition *ne*. Because roles are visible, word order is freer: *ānkhen Isa Masih ne kholiin* is also grammatical (with different emphasis), while "The eyes Christ opened" is marked in English.
- **Agreement with the object.** After *ne* the verb agrees with the *object*: *ānkhen* is feminine plural, so the verb is *kholiin*; for a masculine singular object it is *khola* (*Isa Masih ne darwaza khola*, "Christ opened the door"). English verbs show no such agreement.
- **No articles; postpositions.** Hindi has no *the / a* (definiteness comes from context), and English *in the temple* (preposition + NP) corresponds to *mandir mein* (NP + postposition).

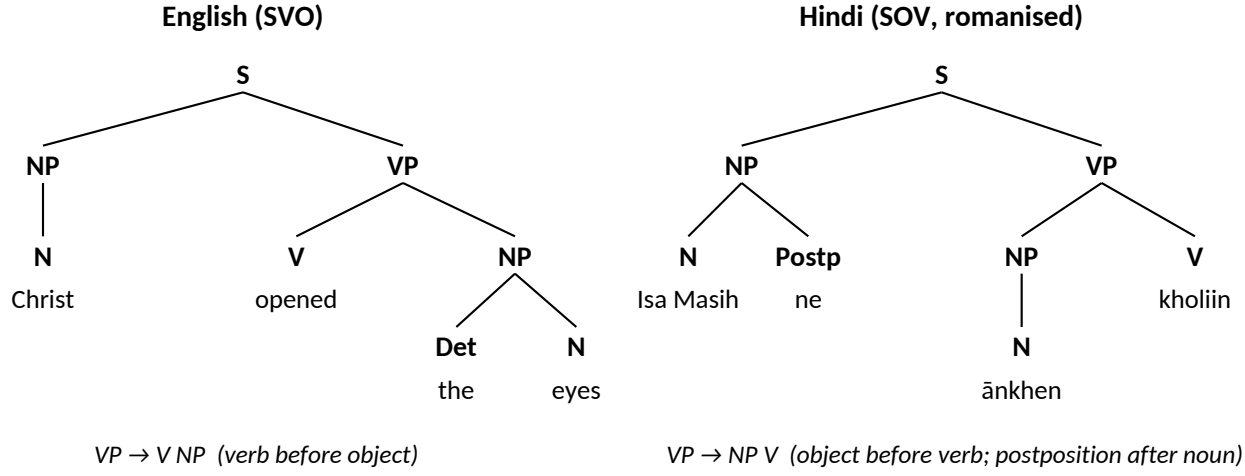


Figure 2: *Phrase-structure trees for "Christ opened the eyes" in English and in Hindi (romanised).*

Table 3: Summary of the differences

Feature	English	Hindi
Basic order	Subject – Verb – Object	Subject – Object – Verb
Adpositions	Prepositions (*in the temple*)	Postpositions (*mandir mein*)
Roles marked by	Word position	Case markers (*ne*, *ko*, *se*) plus position
Word-order freedom	Rigid	Fairly free (scrambling)
Verb agreement	Tense only (*opened*)	Gender, number, person; may agree with the object
Articles	*the*, *a* (obligatory)	None

(b) How a machine translation (MT) system can deal with the difference

English→Hindi translation must **reorder** constituents, **insert or delete** function words and **generate agreement**. Each family of MT systems does this differently.

1. Rule-based transfer MT (analysis → transfer → generation)

- **Analysis:** [S [NP Christ] [VP [V opened] [NP the eyes]]], tense = past, transitive.

- **Lexical transfer:** Christ → *Isa Masih*, open → *khol-*, eye → *ānkh* (feminine).
- **Structural transfer:** rule VP[V NP] → VP[NP V] moves the verb after the object, and *the* is deleted because Hindi has no article.
- **Insertion and agreement:** the rule "past transitive → mark the subject with *ne*" inserts the postposition, and the generator copies gender and number from the object to the verb: *Isa Masih ne ānkhen kholiin.*

The reverse direction (Hindi→English) undoes these steps and must also **insert** an article, guessed from context.

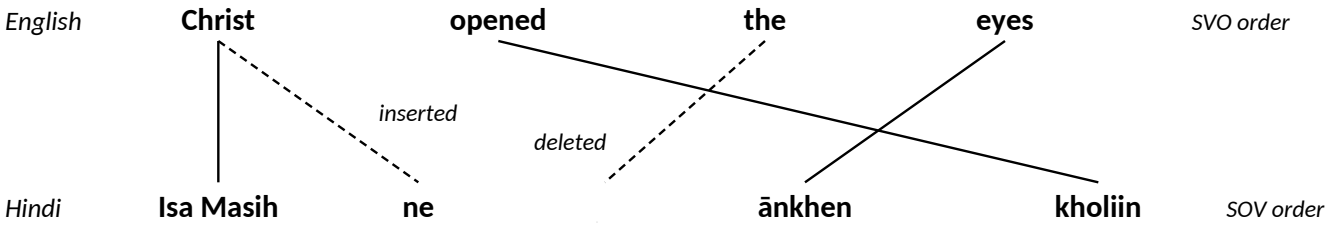


Figure 3: *Word alignment for the transfer example. Crossing lines show reordering, the dashed arrows show an inserted word (ne) and a deleted word (the).*

2. Statistical MT

- Phrase-based SMT learns phrase pairs such as *the eyes* ↔ *ānkhen* from word-aligned parallel text. Long-distance verb movement is hard, so it uses a **distortion / lexicalised reordering model** and a target language model that prefers verb-final Hindi.
- **Pre-ordering:** the English input is parsed and rearranged into Hindi-like order ("Christ the eyes opened") before translation, so alignment is almost monotone. **Syntax-based SMT** (e.g. hierarchical rule $X \rightarrow \langle V\ X_1,\ X_1\ V \rangle$) builds the reordering into the grammar.

3. Neural MT (encoder–decoder with attention / Transformer)

- Self-attention lets the decoder look at any source position, so the verb can be generated last without explicit rules; the model learns SVO→SOV reordering, the insertion of *ne* and gender/number agreement from data, with subword (BPE) units for Hindi morphology. Errors remain for rare words and long sentences.

Evaluation and challenges. Translation quality is judged with automatic metrics such as BLEU (n-gram overlap with a reference translation) and with human ratings of adequacy and fluency. Word-order mistakes, for example leaving the verb in its English position, lower both scores. Remaining difficulties are ambiguous words, names and transliteration (*Christ* → *Isa Masih*), and language pairs with little parallel data.

Conclusion. The languages differ in word order, adpositions, case marking, agreement and articles. Transfer systems use explicit rules, statistical systems use reordering and syntax-based models, and neural systems learn all of it from data, giving *Isa Masih ne ānkhen kholiin*.